Free field theory

$$Z = \int D\varphi \; e^{\frac{i}{\hbar}\int d^4x \left\{ \left(\frac{c^2}{2} \partial_\mu \varphi \partial^\mu \varphi - \frac{1}{2} m^2 \varphi^2 \right) + J\varphi \right\}}$$

$$= \int D\varphi \; e^{\frac{i}{\hbar}\int dx \, \varphi \left\{ -\frac{1}{2} c^2 (\partial^2 + m^2) \right\} \varphi + J\varphi}$$

$$\varphi(x) = \varphi(t, \vec{r}) = \int \frac{d\omega}{2\pi} \int \frac{d\vec{k}}{(2\pi)^3} \; e^{i(\vec{k}\vec{x} - \omega t)} \varphi(\omega, \vec{k})$$

Then $\int dt \, d\vec{r} \; \frac{1}{2} \left(\frac{\partial \varphi}{\partial t} \right)^2 - \frac{c^2}{2} \left(\frac{\partial \varphi}{\partial \vec{x}} \right)^2 =$

$$= \int dt \, d^3\vec{r} \; \frac{1}{2} \int \frac{d\omega \, d\vec{k}}{(2\pi)^4} \int \frac{d\omega' \, d\vec{k}'}{(2\pi)^4} \left\{ (-i\omega)(-i\omega') \, \varphi(\omega, \vec{k}) \varphi(\omega', \vec{k}') \right.$$

$$\left. - c^2 (i\vec{k})(i\vec{k}') \, \varphi(\omega, \vec{k}) \varphi(\omega', \vec{k}') \right\} e^{i\vec{r}(\vec{k} + \vec{k}') - it(\omega + \omega')}$$

$$= \left(\vec{k}' = -\vec{k}, \; \omega' = -\omega \right) = \int \frac{d\omega \, d\vec{k}}{(2\pi)^4} \; \frac{1}{2} (\omega^2 - c^2 k^2) \, \varphi(\omega, \vec{k}) \varphi(-\omega, -\vec{k})$$

$$Z = \int D\varphi(\omega, \vec{k}) \; \exp \frac{i}{\hbar} \int \frac{d\omega \, d\vec{k}}{(2\pi)^4} \left\{ \frac{1}{2} (\omega^2 - c^2 k^2 \overset{-m^2}{\checkmark}) \, |\varphi(\omega, \vec{k})|^2 + \right.$$

$$\left. + \frac{1}{2} J(\omega, \vec{k}) \, \varphi(-\omega, -\vec{k}) + \frac{1}{2} J(-\omega, -\vec{k}) \, \varphi(\omega, \vec{k}) \right\}$$

$$= \int D\varphi \; \exp \frac{i}{\hbar} \int \frac{d\omega \, d\vec{k}}{(2\pi)^4} \; \varphi(\omega, \vec{k}) \; \frac{1}{2} (\omega^2 - c^2 k^2 - m^2 + i\varepsilon) \, \varphi(-\omega, -\vec{k})$$

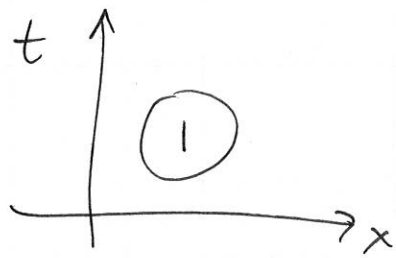
$$\exp -\frac{i}{2\hbar} \int \frac{d\omega \, d\vec{k}}{(2\pi)^4} \; J(-\omega, -\vec{k}) \; \frac{1}{\omega^2 - c^2 k^2 - m^2 + i\varepsilon} \; J(\omega, \vec{k})$$

Using 4-vector $k^i = (\omega, c\vec{k})$

$$Z(J) = Z(0)\, e^{\frac{i}{\hbar}\left(-\frac{1}{2}\int \frac{d^4k}{(2\pi)^4}\, J^*(k)\, \frac{1}{k^2-m^2+i\varepsilon}\, J(k)\right)}$$

let $J(x) = J_1(x) + J_2(x)$ where sources $J_{1,2}(x)$ are concentrated in local regions 1, 2.

~~Local sources~~
~~$J_i(x) = \delta(x - x_i)$~~



$$J(k) = J(\omega, \vec{k}) = \int dt\, d^3\vec{r}\; e^{i(\omega t - \vec{k}\vec{r})}\, J(t, \vec{r})$$

Then $W = -\frac{1}{2}\int \frac{d\omega\, d\vec{k}}{(2\pi)^4} \int dt\, d^3\vec{r} \int dt'\, d^3\vec{r}\,' \; \frac{e^{i\omega(t-t') - i\vec{k}(\vec{r}-\vec{r}\,')}}{\omega^2 - c^2k^2 - m^2 + i\varepsilon}\, J(t,\vec{r})\, J(t', \vec{r}\,')$

$$= -\frac{1}{2}\int dt\, d^3\vec{r} \int dt'\, d^3\vec{r}\,' \underbrace{J(x)}_{J(t,\vec{r})}\; G(t-t', \vec{r}-\vec{r}\,')\, \underbrace{J(x')}_{J(t',\vec{r}\,')}$$

let $J(t, \vec{r}) = \underbrace{\delta(\vec{r} - \vec{r}_1)}_{J_1} + \underbrace{\delta(\vec{r} - \vec{r}_2)}_{J_2}$ (t-independent)

Then we'll get $\sim \int dt\, dr \int dt'\, dr' \left(J_1 J_1 + J_2 J_2 + J_1 J_2 + J_2 J_1\right)$

Keep only cross-terms $J_1 J_2 + J_2 J_1$

$$W = -\frac{1}{2}\int \frac{d\omega\, d\vec{k}}{(2\pi)^4} \int dt\, d^3\vec{x} \int dt\, d^3\vec{y}\,' \; \frac{e^{i\omega(t-t') - i\vec{k}(\vec{x}-\vec{y}\,')}}{\omega^2 - c^2k^2 - m^2 + i\varepsilon} \cdot$$

$$\cdot\, \cancel{\delta(\vec{x} - \vec{x}_1)\, \delta(\vec{y}\,' - \vec{y}_1)} \cdot \left(\delta(\vec{r} - \vec{r}_1)\, \delta(\vec{r}\,' - \vec{r}_2) + \delta(\vec{r} - \vec{r}_2)\, \delta(\vec{r}\,' - \vec{r}_1)\right)$$

$$W = -\frac{1}{2}\int dt\, dt' \int \frac{d\omega\, d^3\vec{k}}{(2\pi)^4} \frac{e^{i\omega(t-t')}}{\omega^2 - c^2k^2 - m^2 + i\varepsilon} \left(e^{-i\vec{k}(\vec{r}_1-\vec{r}_2)} + e^{-i\vec{k}(\vec{r}_2-\vec{r}_1)} \right)$$

$= (\vec{k} \to -\vec{k}$ in the 2nd term$)$

$$= -\frac{2}{2}\int dt\, dt' \int_{(\omega,\vec{k})} e^{i\omega(t-t')} \frac{e^{i\vec{k}(\vec{r}_1-\vec{r}_2)}}{\omega^2 - c^2k^2 - m^2 + i\varepsilon} = \left(\int dt'\, e^{-i\omega t'} = 2\pi\delta(\omega) \right)$$

$$= +\frac{2}{2}\underbrace{\int dt}_{T} \int \frac{d\vec{k}}{(2\pi)^3} \frac{e^{i\vec{k}(\vec{r}_1-\vec{r}_2)}}{c^2k^2 + m^2}$$

total

But remember $Z(J) = Z(0)\, e^{\frac{i}{\hbar}W}$ stands for vacuum-to-vacuum transition amplitude $\langle 0| e^{-i\hat{H}T/\hbar} |0\rangle = e^{-\frac{i}{\hbar}ET}$

where E is the energy due to sources

$\Rightarrow$ set $W = -ET$ and find

$$E = -\int \frac{d\vec{k}}{(2\pi)^3} \frac{e^{i\vec{k}(\vec{r}_1-\vec{r}_2)}}{c^2k^2 + m^2}$$

$$E = -\frac{1}{(2\pi)^2 r c^2}\, \pi e^{-mr/c} = -\frac{1}{4\pi c^2 r} e^{-mr/c} \xrightarrow[m\to 0]{} \frac{-1}{4\pi c^2 r}$$

(Yukawa potential)

$$\frac{dE}{dr} = \frac{1}{4\pi c^2}\left(\frac{m}{cr} + \frac{1}{r^2}\right) e^{-\frac{mr}{c}} > 0$$

the sources attract each other!

$$I_1 = \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta \int_0^{\infty} \frac{dk\, k^2}{(2\pi)^3} \frac{e^{ikr\cos\theta}}{c^2k^2+m^2} =$$

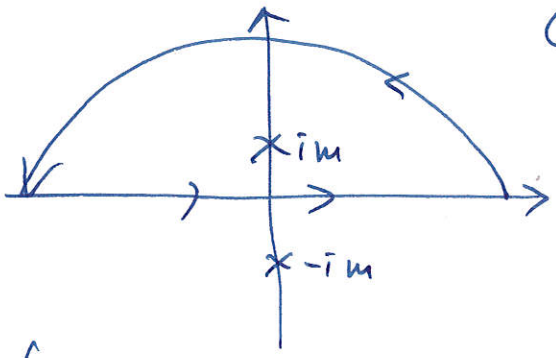
$$= \frac{1}{(2\pi)^2} \int_{-1}^{1} d(\cos\theta) \int_0^{\infty} dk\, k^2 \frac{e^{ikr\cos\theta}}{c^2k^2+m^2} =$$

$$= \frac{1}{(2\pi)^2} \int dk\, k^2 \frac{1}{c^2k^2+m^2} \frac{1}{ikr} e^{ikr\cos\theta}\Big|_{-1}^{1}$$

$$= \frac{1}{(2\pi)^2} \int_0^{\infty} dk\, k^2 \frac{2i\sin(kr)}{ikr\,(c^2k^2+m^2)}$$

$$= \frac{2}{(2\pi)^2 r} \underbrace{\int_0^{\infty} dk \frac{k\sin(kr)}{c^2k^2+m^2}}_{\frac{1}{2}\int_{-\infty}^{\infty} dk \frac{k\sin(kr)}{c^2k^2+m^2}} = \frac{2}{(2\pi)^2 r c^2} \frac{1}{2} I_2$$

$$\Rightarrow I_2 = \int_{-\infty}^{\infty} du \frac{u\sin(ur/c)}{u^2+m^2} = \mathrm{Im} \int_{-\infty}^{\infty} du \frac{u\, e^{iur/c}}{u^2+m^2}$$



$\frac{r}{c} > 0$

$$I_2 = \mathrm{Im}\left\{ 2\pi i \frac{im}{2im} e^{i(im)\frac{r}{c}} \right\} = \pi e^{-mr/c}$$

Evaluate free propagator